PRACTICAL COURSE WORKBOOK

Database Design & Modelling

Model, query and explain dependable data

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	ERD, Normalisation, Schemas, Relationships, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use ERD appropriately in a realistic, bounded task.
- Use Normalisation appropriately in a realistic, bounded task.
- Use Schemas appropriately in a realistic, bounded task.
- Use Relationships appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Database Design & Modelling is a structured, practical course covering ERD, Normalisation, Schemas, Relationships. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 ERD: Data modelling	1. Entities with ERD 2. Relationships with Normalisation 3. Constraints with Schemas
02 Normalisation: Querying	1. Selection with Relationships 2. Joins with ERD 3. Aggregation with Normalisation
03 Schemas: Reliability	1. Transactions with Schemas 2. Indexes with Relationships 3. Backups with ERD
04 Relationships: Production design	1. Security with Normalisation 2. Performance with Schemas 3. Operations with Relationships

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

ERD: Data modelling

Apply ERD through entities, relationships, constraints.

LESSON 01 / 18 MIN

Entities with ERD

Learning objectives

- Explain entities in the context of Database Design & Modelling.
- Apply ERD to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

entities, ERD, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a entities result produced with ERD?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Relationships with Normalisation

Learning objectives

- Explain relationships in the context of Database Design & Modelling.
- Apply Normalisation to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

relationships, Normalisation, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a relationships result produced with Normalisation?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Constraints with Schemas

Learning objectives

- Explain constraints in the context of Database Design & Modelling.
- Apply Schemas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

constraints, Schemas, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a constraints result produced with Schemas?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Normalisation: Querying

Apply Normalisation through selection, joins, aggregation.

LESSON 04 / 30 MIN

Selection with Relationships

Learning objectives

- Explain selection in the context of Database Design & Modelling.
- Apply Relationships to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

selection, Relationships, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a selection result produced with Relationships?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Joins with ERD

Learning objectives

- Explain joins in the context of Database Design & Modelling.
- Apply ERD to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

joins, ERD, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a joins result produced with ERD?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Aggregation with Normalisation

Learning objectives

- Explain aggregation in the context of Database Design & Modelling.
- Apply Normalisation to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

aggregation, Normalisation, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a aggregation result produced with Normalisation?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Schemas: Reliability

Apply Schemas through transactions, indexes, backups.

LESSON 07 / 26 MIN

Transactions with Schemas

Learning objectives

- Explain transactions in the context of Database Design & Modelling.
- Apply Schemas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transactions, Schemas, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a transactions result produced with Schemas?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Indexes with Relationships

Learning objectives

- Explain indexes in the context of Database Design & Modelling.
- Apply Relationships to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

indexes, Relationships, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a indexes result produced with Relationships?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Backups with ERD

Learning objectives

- Explain backups in the context of Database Design & Modelling.
- Apply ERD to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

backups, ERD, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a backups result produced with ERD?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Relationships: Production design

Apply Relationships through security, performance, operations.

LESSON 10 / 22 MIN

Security with Normalisation

Learning objectives

- Explain security in the context of Database Design & Modelling.
- Apply Normalisation to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Normalisation, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a security result produced with Normalisation?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Performance with Schemas

Learning objectives

- Explain performance in the context of Database Design & Modelling.
- Apply Schemas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

performance, Schemas, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a performance result produced with Schemas?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Operations with Relationships

Learning objectives

- Explain operations in the context of Database Design & Modelling.
- Apply Relationships to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

operations, Relationships, databases

Retrieval prompt

In Database Design & Modelling, which evidence best supports a operations result produced with Relationships?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Database Design & Modelling project using ERD, Normalisation, Schemas.

DELIVERABLE	A working Database Design & Modelling artefact plus an evidence-based self-review.
TOOLS	A suitable ERD environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using ERD.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE
Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE
What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION
What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21

https://www.postgresql.org/docs/current/tutorial.html

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21

https://www.postgresql.org/docs/current/sql.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the ERD scope.